MODULE :3

Electrical Properties of Metals and Semiconductors

Introduction: Since metals are good conductors of electricity and it is consider that electrons can move freely inside the metal. These electrons respond to an external applied electric field which produces current. And this consideration of free electrons is known as free electron theory of metals. Later this theory was modified and called as classical free electron theory and used to explain many properties. Some important terminologies related are

Mean free path (λ): It is the average distance covered by the free electron between two successive collisions and is represented as λ.

Mean Collision Time (τ):

It is the average time that elapses between two consecutive collisions of an electron with thelattice ions.

Drift Velocity ($\boldsymbol{Vd}$):It the average velocity with which the free electron moves along the positive potential and represented by$\boldsymbol{Vd}$

Resistivity:

Resistivity is a fundamental property of materials that quantifies how strongly they resist the flow of electric current. It is often denoted by the Greek letter ρ (rho)

Resistivity (ρ) is defined as the resistance of a uniform material sample with a specific geometry of unit length and unit cross-sectional area.. It is given by

$$\rho = R \frac{A}{L}$$

Where, R→Resistance of the material

A→Cross-sectional area of the material,

L→ Length of the material

Mobility:

Mobility can be defined as the ratio of magnitude of drift velocity to the electric field. It is given by

$$\mu = \frac{|V_d|}{E}$$

Where, μ– mobility of the charge carrier

V_d–drift velocity of charge carriers

E– external applied electric field

Relaxation time (τ_r):

Relaxation time is the time required for the average velocity of free electrons to reduce to (1/e) times its value that existed when the field is just turned off.

Expression for electrical conductivity:

As per the classical free electron theory, the conductivity of conductor is given by

$\sigma=ne^2\tau/m$

where,σ–electrical conductivity

n–free electron concentration,

τ–mean collision time

e–charge on the electron,

m–mass of the electron

Electron Scattering:

1. **Electron–Electron Scattering:**

   Electron-electron scattering is the deflection of one electron's path due to interaction with another, driven by electrostatic forces. This can be viewed as a process in which electrons are deflected when passing through a material or when colliding with each. It is a fundamental process that influences material properties like electrical conductivity.

2. **Electron-Lattice Scattering:**

   Electron lattice scattering is the process where electrons are deflected from their path by the vibrations of atoms in a crystal lattice, also known as phonon scattering. This occurs because thermal energy causes atoms to vibrate around their positions. The rate of scattering increases with temperature

3. **Electron-Impurity Scattering:**

   Electron impurity scattering is the process where an electron's path is deflected by the electrostatic force of ionized impurities in a crystal. This scattering reduces electron mobility, and the effect is more significant at lower temperatures and higher doping concentrations.

4. **Electron–Defects Scattering:**

   Electron defect scattering is the deflection of electrons as they travel through a material due to interactions with imperfections, or "defects," in the crystal lattice, such as vacancies, dislocations, or impurities. These scattering events alter the electron's path and affect the material's overall properties, including electrical conductivity and thermoelectric performance.

: Failures/Drawbacks of classical free electron theory:

Although electrical and thermal conductivity in metals can be explained successfully throughclassical free electron theory, it failed to account for many other experimental

facts such as Specific heat, temperature dependence of ' σ '& dependence of electrical conductivity on electron concentration

1. **Specific heat**:

The molar specific heat of gas molecules at constant volume is $C_v = \frac{3}{2}R$, where R is a universal constant.

But the experimental value of specific heat for conduction electrons is found to be $C_v=10^{-4}RT$ which the classical theory could not explain.

Also the experimental value shows that the specific heat value is low and temperature dependent, whereas the classical free electron theory says that it is temperature independent.

Hence the classical free electron theory fails to explain the above facts hence it fails.

2.Temperature dependence of electrical conductivity:

The electrical conductivity of a metal is inversely proportional to temperature.

$$\sigma \ \alpha \ \frac{1}{T} \text{ -------1}$$

According to classical free electron theory, electrical conductivity is inversely proportional to the square root of temperature, i.e.

$$\sigma \ \alpha \ \frac{1}{\sqrt{T}}$$

$$\frac{3}{2}kT = \frac{1}{2}mv_{th}^2$$

$$V_{th} = \sqrt{\frac{3kT}{3}}$$

$$V_{th} \ \alpha \ \sqrt{T}$$

$$\tau \ \alpha \ \frac{1}{v}$$

$$\sigma = \frac{ne^2\tau}{m}$$

$$\sigma \ \alpha \ \frac{1}{\sqrt{T}} \text{ ------2}$$

Hence the classical free electron theory fails to explain the above ie 1 and 2 is not agreed hence it fails.

3.Dependence of electron concentration with conductivity:

According to classical electron theory electrical conductivity is directly proportional to the electron concentration. But monovalent metals like copper found to have high electrical conductivity than the divalent & trivalent metals like Zinc and Aluminium. Hence CFET fails to explain the observation.

Consider zinc and cadmium which are divalent metals. Their electrical conductivities are $1.09\times10^7/\Omega$m and $0.15\text{x}10^7/\Omega$m. These are much lesser than that of monovalent metals copper and silver. The values of which are $5.88\text{x}10^7/\Omega$m and $6.3\text{x}10^7/\Omega$m respectively.

The electron concentrations for zinc and cadmium are $13.1\times10^{28}/\text{m}^3$ and $9.28\times10^{28}/\text{m}^3$

which are much higher than that for copper and silver, the values of which are $8.45\times10^{28}/m^3$ and $5.85\times10^{28}/m^3$ respectively.

Hence the classical free electron theory fails to explain the dependence of σ on electron concentration.

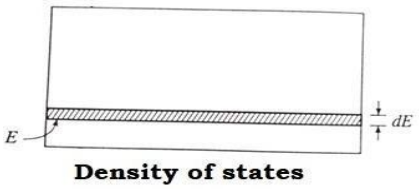
Assumptions of quantum free electron theory:

The main assumptions of quantum free electron theory are

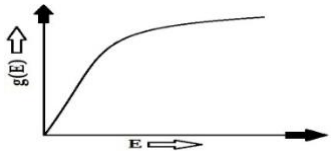
1. The energy values of free electrons are quantized. The allowed energy values are realized interms of a set of energy levels.
2. The distribution of electrons in the various allowed energy levels follows Pauli's exclusion principle.
3. The free electrons travel in a constant potential inside the metal but stay confined within its boundaries.
4. The attraction between the free electrons and lattice ions, the repulsion between the electronsthemselves are ignored.

Density of states g (E):

The permitted energy levels for electrons in a solid will be in terms of bands. Each energy band



Density of states

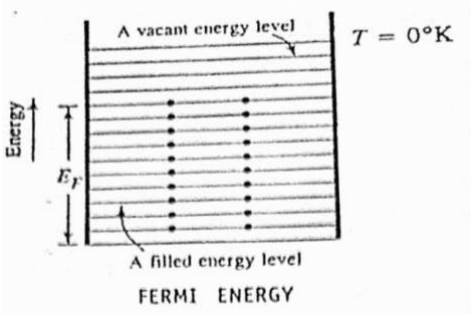


spread over an energy range of few eV. The number of energy levels in each band will be extremely large and hence the energy values appear to be virtually continuous over the band spread. Each energy level consists of two states and each state accommodates only one electron. Therefore, energy level can be occupied by two electrons only, having opposite directions of spin. The exact dependence of density of energy states on the energy is realized through a function denoted as g (E) and is known as density of states function. It is defined as, "**the number of available states per unit volume per unit energy interval"**. The number of states lying in the range of energies between E and E+dE is given by

$$g(E) = \frac{8\sqrt{2}\,\pi m^{\frac{3}{2}}}{h^3} E^{\frac{1}{2}}$$

Fermi Energy:

In a metal having N atoms, there are N allowed energy levels in each band. In the energy band the energy levels are separated by energy differences. It is characteristics of the material. According to Pauli's exclusion principle, each energy level can accommodated a maximum of two electrons, one with spin up and spin down. The filling up of energy levels occurs from the lowest level. The next pair of electrons occupies the next allowed energy levels and so on till all the electrons in the metal are accommodated. Still number of allowed energy levels is left vacant. This is the picture when there is no external energy supply for the electrons. "**The energy corresponding to the highest occupied level by electrons at absolute zero temperature (0K) is called Fermi energy and the energy level is called Fermi level"**. It is denoted by E_F. At 0 Kelvin, all states below the Fermi energy are filled, and all states above are empty.



Fermi Factor:

When the temperature is not at absolute zero, the material will be receiving thermal energy from surroundings. At ordinary temperature, electrons near the Fermi Level are capable of being excited. During thermal excitation (i.e., T >0), the electrons which absorb the thermal energy move into higher energy levels which were unoccupied at zero degree absolute (i.e., T= 0). Though such excitations seem to be random, the occupation of various energy level obey a statistical distribution called Fermi-Dirac distribution, once the system is in thermal equilibrium (i.e., at a steady temperature state).

"**The probability f (E) that a given energy state with energy E is occupied at a steady temperature T"** is given by

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}$$

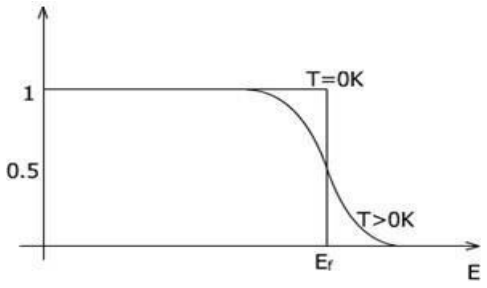
Fermi-Dirac statistics:

Under thermal equilibrium the free electrons are distributed in various available energy states. The distribution of electrons among the energy levels follows statistical rule known as Fermi Dirac statistics. Fermi-Dirac statistics is applicable to fermions. Fermions are indistinguishable particles with half integral spin. Since electron has half spin they obey Fermi-Dirac statistics and they are called Fermions. Fermi factor represents the probability that a quantum state with energy E is occupied by an electron, is given by Fermi-Dirac distribution function,

$$F(E) = \frac{1}{e^{\frac{E-E_F}{kT}}+1}$$

Where k is the Boltzmann's constant, T is the temperature in Kelvin, E is the energy and E_F is the Fermi energy.

Dependence of Fermi factor on temperature & Energy/ Variation of Fermi factor with Temperature & Energy.



The dependence of Fermi factor on temperature at T=0K is given in the figure.

Probability of occupation is considered for following cases,

Case 1: Probability of occupation for E < EF at T = 0K

Substituting the value of T = 0K in the Fermi function we get

$$F(E) = \frac{1}{e^{\frac{E-E_F}{kT}}+1}$$

$$F(E) = \frac{1}{e^{-\infty}+1}$$

$$F(E) = 1$$

It means that, the energy levels are certainly occupied i.e., there is 100% probability that the electrons occupy the energy level below Fermi level.

Case 2: Probability of occupation for E > EF at T = 0K

Substituting the value of T = 0K in the Fermi function we get

$$F(E) = \frac{1}{e^{\frac{E-E_F}{kT}}+1}$$

$$F(E) = \frac{1}{e^{\infty}+1} = \frac{1}{\infty} = 0$$

$$F(E) = 0$$

All the energy levels above Fermi level are unoccupied i.e., 0% probability for the electrons to occupy the energy level above the Fermi level.

Case 3: Probability of occupation for E = EF at T = 0K

$$F(E) = \frac{1}{e^{\frac{E-E_F}{kT}}+1}$$

$$F(E) = \frac{1}{2} = 0.5$$

There is 50% probability for the electrons to occupy the Fermi energy level.

Electrical Conductivity in the Quantum Free Electron Theory (QFET)

In Quantum Free Electron Theory, the expression for electrical conductivity is derived using the Fermi–Dirac distribution instead of the classical Maxwell–Boltzmann statistics. The conductivity of a metal is given by

$$\boldsymbol{\sigma = \frac{ne^2\,\tau}{m^*} = \frac{ne^2\,\lambda}{m^* v_f}}$$

Where m ∗ is the effective mass of the electron,

Success / Merits of Quantum free electron theory:

(1) Specific heat:

According to classical free electron theory, all the electrons absorb the heat irrespective of energy levels occupied by free electrons, but according to quantum free electron theory, only few electrons near to Fermi level are capable of absorbing the heat energy. Hence the specific heat is given by quantum theory that $C_V=\left(\frac{2k}{E_f}\right)RT$ (or) 10^{-4} RT which agree well with experimental values.

(2) **Temperature dependence of Electrical conductivity**:

We know that electrical conductivity is

$\sigma = \frac{ne^2\tau}{m}$

But according to quantum free electron theory Relaxation time

$$\tau = \frac{\lambda}{v} \text{ and } \sigma = \frac{ne^2\,\tau}{m^*} = \frac{ne^2\,\lambda}{m^* v_f}$$

If the metal is heated, the ionic core start to vibrate, if r is the amplitude of vibration then the area of influence (or) cross sectional area is $\pi r2$.

We say that $r^2 \propto T$ ---------- (2)

This cross sectional area scatter the electrons, hence $r^2 \propto \frac{1}{\pi r^2}$---------- (3)

Comparing equations (2) & (3) we get

$\lambda \propto \frac{1}{T}$---------- (4)

By comparing equation (1) & (4)

We get $\sigma \propto \frac{1}{T}$ thus dependence of σ on T is correctly explained on the basis of quantum free electron theory.

(3) **Electrical conductivity and electron concentration** :

We know that electrical conductivity $\sigma = \frac{ne^2\,\tau}{m^*} = \frac{ne^2\,\lambda}{m^* v_f}$ So that electrical conductivity is

depend on both concentration of electron (n) as well as $\frac{\lambda}{v}$ also. It explains the existence of long mean free path at low temperature from the following relation $\lambda \propto \frac{1}{T}$

Quantum free electron theory explains successfully electrical conductivity & thermal conductivity. It explains paramagnetic susceptibility and heat capacity successfully.